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Ultra-broadband dual-branch optical frequency comb with 10^{-18} instability

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Coherently bridging across wide spectral regions in the optical domain with ultra-high precision will enable progress in optical spectroscopy and time and frequency metrology. We report optical frequency synthesis at the 10^{-18} instability level based on a novel technique actively canceling the differential phase noise of a dual-branch Er-fiber based optical frequency comb spanning 500–2200 nm. We validate the method with an out-of-loop measurement demonstrating the stability transfer from 1560 nm to 780 nm with an instability level at 3×10^{-18} at 1 s averaging time and 1×10^{-19} at 1000 s (approximately averaging on $1 \times 10^{-17}/\sqrt{\tau}$), limited mainly by uncompensated optical interferometers used to detect optical beatnotes. To confirm this limitation, we propose an experimental setup that eliminates the excess noise induced by uncompensated optical paths and show that the proposed method exhibits an ultimate unprecedented instability level at 6×10^{-19} at 1 s averaging time when two optical signals can share a common photodetector. The system realized becomes a powerful and universal tool for optical frequency comparison and spectral purity transfer across the visible and infrared domains.

OCIS codes: (120.3940) Metrology; (140.4050) Mode-locked lasers; (140.7090) Ultrafast lasers; (320.6629) Supercontinuum generation; (140.3515) Lasers, frequency doubled.

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1. INTRODUCTION

The stability and accuracy of optical atomic clocks [1–6] grant quantum jumps in test of fundamental physics [7–11], geodesy [12,13], detection of dark matter [14,15], gravitational wave detection [16], and the realization of an optical timescale [17]. They are typically limited by the stability of the laser used to excite the atomic transition. Improvements are being sought through novel approaches to laser frequency stabilization [18–24], but these are often restricted to certain wavelengths. An optical frequency comb must therefore be used to transfer the stability to the desired wavelength(s), with a resultant compromise between the stability transfer performance achievable and the number of wavelengths that can be accessed simultaneously.

Optical frequency combs are essential in providing a straightforward means for comparing atomic clocks with disparate transition frequencies. [25]. They provide a convenient way of phase-coherently linking clocks operating at different frequencies, allowing both optical frequency ratios and absolute optical frequencies relative to cesium primary standards to be measured. The accuracy of such measurements has been shown to be limited by the accuracies of the frequency standards involved rather than by the frequency comparison technique [26–28]. Optical frequency combs also enable the stability of an optical reference to be transferred to other frequencies, either in the optical, THz, or microwave domains [29–34].

The transfer of frequency stability across the optical spectrum is becoming increasingly important as new generations of ultra-stable lasers are developed in an effort to reduce the instability of optical clocks. For example, state-of-the-art lasers stabilized to cryogenic silicon resonators have an estimated thermal-noise-limited instability floor of 4×10^{-17} [35]. However, because the optical transmission bandwidth of silicon is restricted to the infrared (IR) region between 1 μm and 6 μm, most optical clock transitions are not directly accessible and typically a ~1.5 μm laser is stabilized to the cavity instead. This wavelength is well matched to fiber-based femtosecond comb technology, which is also attractive in terms of achieving robust long-term operation.

In pioneering works, a fiber-based optical frequency comb was used to transfer the stability of a master laser to a slave laser in the IR domain [31,36,37], demonstrating that the excess instability introduced by the comb could be as low as 4×10^{-18} at 1 s averaging time. However, this level of stability was achieved by using a single spectrally broadened comb output to detect the beatnote with two continuous-wave (CW) lasers and operating the comb in the narrow linewidth regime by locking its repetition rate f_{rep} to the master laser. This approach is extremely challenging to extend to multiple wavelengths due to the difficulty of optimizing the broadened spectrum for simultaneous detection of multiple optical beat signals. To enable independent optimization of several beat signals, a multi-branch configuration is usually preferred, in which a dedicated optical amplifier and nonlinear fiber is used to generate a relatively narrow-band frequency comb output around each CW laser wavelength of interest. However, the increased flexibility of this multi-branch approach comes at the expense of differential phase noise between the branches. This has been reported in previous experiments to limit the stability transfer to typically a few parts in 10^{16} level for an averaging time of 1 s [38,39]. Recently, a research team in Japan has demonstrated that it is possible to correlate the accumulated phase noise from one branch to another by using an additional CW laser locked to the comb to detect differential phase noise and feeding it back to an actuator in the optical branch [40]. The servo-locking bandwidth limit makes it difficult to apply enough gain to reach an instability at the 10^{-18} level at 1 s averaging time. Moreover, this technique does not show any trivial way to access visible wavelengths. Periodically poled lithium niobate (PPLN)-based technology can be used to double the output of a single-branch spectrum and to facilitate a measurement of the carrier envelope offset frequency as well as generate a broadband near-IR output [41]. But to date, the achievable signal-to-noise ratio (SNR) has not been sufficient for precision optical frequency metrology.

Here, we propose a novel technique to leverage all those limitations. By interfering, at 1 μm, a visible and an IR branch generated by a common Er-fiber laser, we can detect and stabilize f_{ceo} while simultaneously correlating differential phase noise between the two branches. The detection of f_{ceo} through the IR branch allows detecting the remaining noise in both branches. We then passively cancel the noise by mixing the optical beatnotes with f_{ceo}. An out-of-loop measurement has been performed to demonstrate the stability transfer from 1560 nm to 780 nm can be as low as 3×10^{-18} at 1 s averaging time. We also demonstrate that the ultimate limit of the method we propose is at the 6×10^{-19} instability level at 1 s integration time by eliminating noise of the remaining uncompensated optical paths.

2. FIBER NOISE CANCELLED DUAL-BRANCH FREQUENCY COMB

A. Dual Supercontinuum Generation and Noise Cancellation

The concept of the proposed method and its practical implementation are illustrated in Figs. 1(a) and 1(b). We consider two supercontinua generated by a common femtosecond oscillator operated at 1550 nm. The first supercontinuum is obtained by optical amplification and spectral broadening in a highly nonlinear fiber. Such a supercontinuum will conventionally span 1000–2200 nm. A second supercontinuum is generated by the same method except that the amplified light is doubled to 780 nm prior to spectral broadening. Each supercontinuum is generated in distinct non-common path optical branches. By introducing a PPLN crystal in the IR supercontinuum branch, we can generate the carrier–envelope frequency f_{ceo}. Additionaly, we can detect f_{ceo} from the spectral overlap between the two supercontinua. To mathematically explain the method presented in Figs. 1(a) and 1(b), we consider the phase noises ϕ_A, ϕ_B, ϕ_C, ϕ_D at the four photodetectors A, B, C, and D (for the sake of simplicity, we ignore the noise induced by the radio frequency (RF) references, servo-locking errors, optical interferometers, and photodetectors), respectively. Photodetector A detects the dual-branch f_{ceo} RF signal, photodetector B detects the f'_{ceo} RF signal from the IR branch and PPLN, photodetector C detects the beatnote f_{IR} between an IR CW laser and an IR comb mode generated by the IR branch (through the IR polarization beam splitter IR PBS), and photodetector D detects the beatnote f_{VIS} between a visible CW laser and a comb mode generated by the visible branch (through the visible polarization beam splitter VIS PBS). We consider for both optical beatnotes the case in which the comb mode has a higher frequency than the laser frequency. We write the system of equations as follows:

$$\begin{cases} \phi_A(t) = \phi_{\text{ceo}}(t) + (\phi_{\tau\text{VIS}}(t) - \phi_{\tau\text{IR}}(t)) \\ \phi_B(t) = \phi_{\text{ceo}}(t) + \phi_{\tau\text{IR}}(t) \\ \phi_C(t) = n\phi_{\text{rep}}(t) + \phi_{\text{ceo}}(t) + \phi_{\tau\text{IR}}(t) - \phi_{\nu_{\text{IR}}}(t) \\ \phi_D(t) = m\phi_{\text{rep}}(t) + 2\phi_{\text{ceo}}(t) + \phi_{\tau\text{VIS}}(t) - \phi_{\nu_{\text{VIS}}}(t), \end{cases} \tag{1}$$

with ϕ_{ceo}, the phase noise of the carrier–envelope frequency f_{ceo}, $\phi_{\tau\text{VIS}}$, the phase noise accumulated in the visible branch, $\phi_{\tau\text{IR}}$, the phase noise accumulated in the IR branch, ϕ_{rep}, the phase noise of the repetition rate, $\phi_{\nu_{\text{VIS}}}$, the phase noise of the CW visible laser ν_{VIS}, $\phi_{\nu_{\text{IR}}}$, the phase noise of the CW IR laser ν_{IR}, and n and m are the mode numbers of the comb lines beating with the IR and visible CW lasers, respectively. Conventionally, the carrier–envelope frequency f_{ceo}, generated by the IR branch and the in-line PPLN, is phase locked to a RF reference (which would set $\phi_B = 0$ and hence $\phi_{\text{ceo}} = -\phi_{\tau\text{IR}}$). This approach allows to cancel the noise accumulated in the branch when the two CW lasers are beating with the common branch. Here, we propose, instead, to phase lock the carrier–envelope frequency detected from the dual-branch interferometer to an RF reference. The system of equations then becomes

$$\begin{cases} \phi_A(t) = 0 \rightarrow \phi_{\text{ceo}}(t) = (\phi_{\tau\text{IR}}(t) - \phi_{\tau\text{VIS}}(t)) \\ \phi_B(t) = (2\phi_{\tau\text{IR}}(t) - \phi_{\tau\text{VIS}}(t)) \\ \phi_C(t) = n\phi_{\text{rep}}(t) + (2\phi_{\tau\text{IR}}(t) - \phi_{\tau\text{VIS}}(t)) - \phi_{\nu_{\text{IR}}}(t) \\ \phi_D(t) = m\phi_{\text{rep}}(t) + (2\phi_{\tau\text{IR}}(t) - \phi_{\tau\text{VIS}}(t)) - \phi_{\nu_{\text{VIS}}}(t). \end{cases} \tag{2}$$

From Eq. (2), we write the differential phase noise Δ:

$$\Delta(t) = (2\phi_{\tau\text{IR}}(t) - \phi_{\tau\text{VIS}}(t)). \tag{3}$$

If we phase lock the oscillator repetition rate to the IR CW laser ν_{IR}, the phase noise detected at photodetector D becomes

$$\phi_D(t) = \frac{m}{n}\phi_{\nu_{\text{IR}}}(t) - \phi_{\nu_{\text{VIS}}}(t) + \left(1 - \frac{m}{n}\right)\Delta(t). \tag{4}$$

Equation (4) reveals that there is remaining fiber differential phase noise, as practically the mode numbers are not equal. However, we can observe in the system of Eq. (2) that $\phi_B = \Delta$. We can then remove the phase noise Δ from ϕ_C and ϕ_D. We feed forward the differential phase noise Δ in the optical domain through an acousto-optic modulator or in the RF domain with an RF mixer. We then obtain the final set of equations:

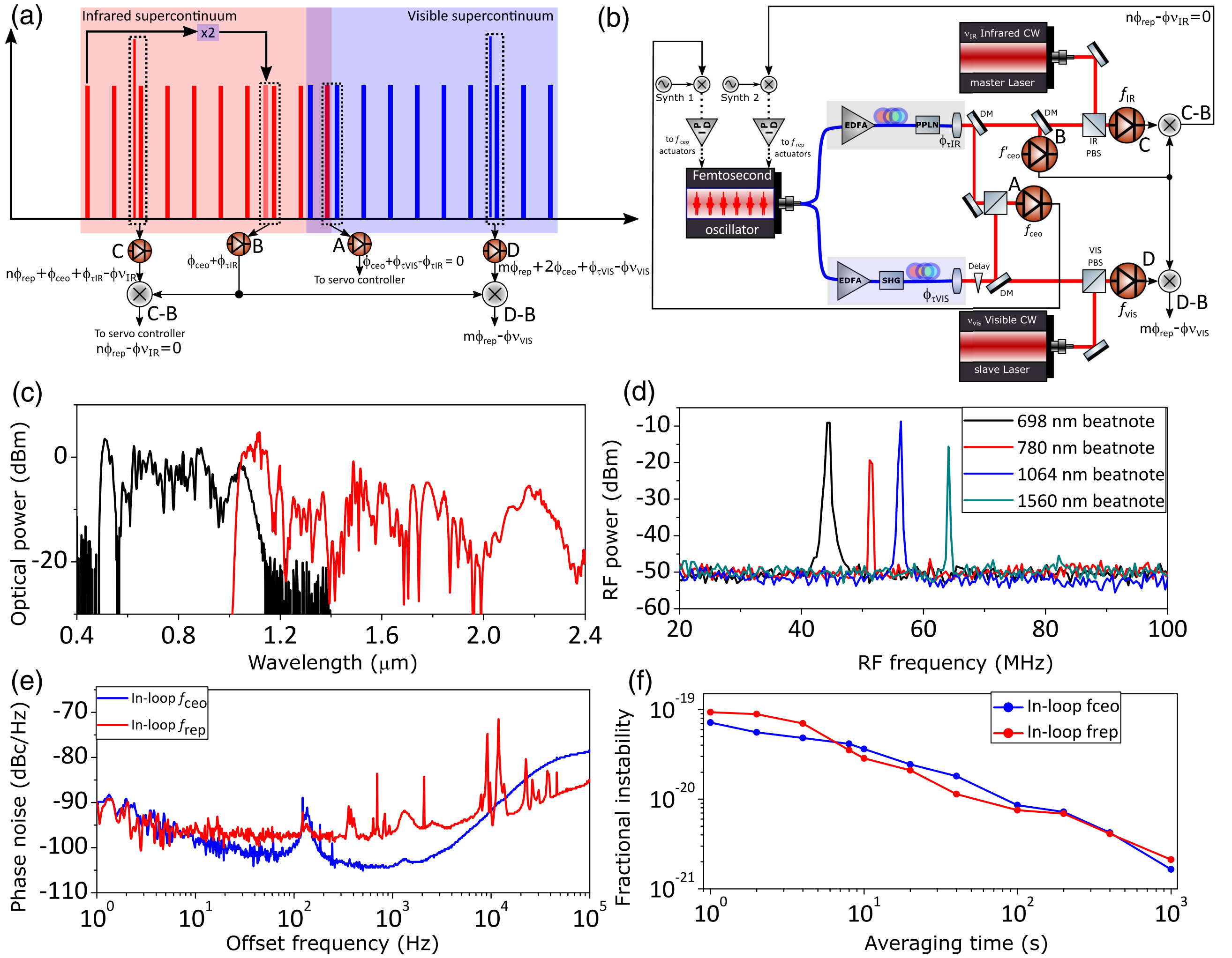

Fig. 1. (a) Representation of the proposed approach to compare two optical frequencies separated by more than an octave in a fiber noise cancelled dual-branch frequency comb. See text for details. (b) Diagram of the proposed method. Two CW lasers are linked through the dual-branch frequency comb with no excessive noise due to non-common optical path. DM, dichroic mirror; EDFA, erbium-doped fiber amplifier; PPLN, periodically poled lithium niobate; SHG, second-harmonic generation. IR PBS, infrared polarization beam splitter; VIS PBS, visible polarization beam splitter. (c) Optical spectra of the dual-branch frequency comb spanning 500 nm to 2200 nm. (d) RF spectra of optical beatnotes at 698 nm (black), 780 nm (red), 1064 nm (blue), and 1560 nm (green), measured with 100 kHz resolution bandwidth in free-running operation. (e) Phase noise power spectral density of the residual error of the servo locking of f_{ceo} (blue) and f_{rep} (red). (f) Fractional frequency instability in terms of Allan deviation of the residual error of the servo locking of f_{ceo} (blue) and f_{rep} (red) (raw data acquired with a Λ-type frequency counter with 1 s gate time).

$$\begin{cases} \phi_C(t) - \phi_B(t) = n\phi_{\text{rep}}(t) - \phi_{\nu_{\text{IR}}}(t) \\ \phi_D(t) - \phi_B(t) = m\phi_{\text{rep}}(t) - \phi_{\nu_{\text{VIS}}}(t). \end{cases} \tag{5}$$

By phase locking the down-mixed RF signal carrying $\phi_C(t) - \phi_B(t)$ to an RF reference with the repetition rate actuators ($\phi_C(t) - \phi_B(t) = 0$), we can write

$$\phi_D(t) - \phi_B(t) = \frac{m}{n}\phi_{\nu_{\text{IR}}}(t) - \phi_{\nu_{\text{VIS}}}(t). \tag{6}$$

The phase noise, accumulated, respectively, in the IR and visible branch, is entirely removed making it possible to compare the stability and phase noise of two CW lasers from different spectral domains without any excess phase noise. Note that this technique can also be used with the transfer oscillator method where direct digital synthesizers would be used to remove the noise of the repetition rate. In the next section, we will describe the implementation of the method illustrated in Fig. 1(b).

B. Supercontinuum Characterization and Beatnote Detection

The 250 MHz Er-fiber oscillator is based on a nonlinear amplifying loop mirror design (IMRA Ecomb-250R [42–44]). The oscillator is split into two distinct branches. The first branch is dedicated to generation of an IR supercontinuum from 1000 nm to 2200 nm, while the other branch generates a visible supercontinuum from 500 nm to 1100 nm (performances in single-branch configuration of the visible supercontinuum are reported in Supplement 1). On the visible branch, the Er:fiber oscillator is amplified with an Er:fiber similariton amplifier optimized for power, pulse width, and amplifier efficiency [45]. The light is then frequency doubled through a PPLN doubling crystal. We then interfere the spectral overlap between the two branches near 1064 nm, producing an f-$2f$ signal, which is utilized for stabilization of f_{ceo} via phase locking to an RF reference.

The in-line PPLN within the IR branch allows the generation of the carrier–envelope frequency at 1100 nm.

The dual output is depicted in Fig. 1(c). If spatially combined, it would give an optical spectrum spanning 500–2200 nm. Note that some spectral regions suffer from low optical power per mode leading to optical beatnotes with low SNR. Nevertheless, this limitation can be pulled through by the utilization of tracking oscillators in the narrow-linewidth regime. Nicolodi *et al.* demonstrated 3×10^{-18} level with only 10 dB SNR in 1 MHz bandwidth [31]. The main drawback in this situation is that the slave lasers would need to be pre-stabilized to linewidth <100 Hz. In this paper, we demonstrate four optical beatnotes detection in different spectral ranges, at 698 nm, 780 nm, 1064 nm, and 1560 nm. The RF signals are shown in Fig. 1(d). The SNRs are higher than 30 dB in 100 kHz resolution bandwidth, which makes it practical to use in the narrow-linewidth regime without the use of tracking oscillators.

C. Residual Error of the Oscillator

In order to control the phase noise and the stability of the carrier–envelope frequency f_{ceo} and the repetition rate f_{rep} through an optical beatnote f_{beat}, we introduce in the oscillator cavity some actuators. For the control of f_{ceo}, we use a fast modulation provided by a graphene electro-optic modulator [46]. The achievable bandwidth of such a modulator can be as high as 1 MHz. For the repetition rate f_{rep}, the fast control is realized by a lithium niobate electro-optic modulator. This allows the modulation bandwidth to be as high as 1 MHz. We use a piezo-electric transducer (PZT) for the slow control. Such high bandwidth actuators allow to apply very high closed-loop gain permitting to correlate the noise of the RF reference and the controlled signals very close to the carrier. The in-loop phase noise of f_{ceo} and f_{beat} (we compare the signals f_{ceo} and f_{beat} with a synthesizer derived from the same reference used to synthesize the RF signal used for the servo-lockings) is shown in Fig. 1(e). We chose to focus on the phase noise for offset frequencies between 1 Hz and 1 kHz. Both signals exhibit phase noise power spectral density (PSD) of −90 dBc/Hz at 1 Hz offset frequency. At 1 kHz offset frequency, both phase noise PSDs can be influenced by each other due to cross talk between the two actuations. However, these in-loop phase noise PSD levels can support the best optical oscillators or optical clocks. Moreover, we show in Fig. 1(f), the Allan deviation (scaled to an optical frequency at 1064 nm for f_{ceo} and 1560 nm for f_{beat}) of the in-loop errors. Both in-loop instability levels are lower than 1×10^{-19} at 1 s averaging time and averaging down to 1×10^{-21} at 1000 s averaging time. To compute these Allan deviations, as well as those throughout the paper, we have used the normal Allan deviation equation with raw data from a Λ-type frequency counter with 1 s gate time (additional information is given in Supplement 1 on how the data are being counted). This in-loop instability is competing with the best laser frequency comb technology [47] thanks to the intrinsically low jitter of the laser in combination with high-bandwidth actuators. The quadratic sum of the in-loop phase noise and stability of f_{ceo} and f_{rep} represents the ultimate limit of a spectral purity transfer, and it is important to mention that this limit should be kept as low as possible.

3. OUT-OF-LOOP CHARACTERIZATION

A. Stability Transfer Demonstration

In the previous section, either for the principle demonstration or the in-loop data, we gloss over unavoidable technical noise and errors. Indeed, while we have quantified the residual errors due to servo-locking, we have not introduced additive noise due to non-common optical paths and interferometers, photodetectors, and electrical filtering and amplification chains. An out-of-loop validation of the new concept is then required. Conventionally, duplicating the experimental setup allows a full characterization of the residual error. However, using a dual frequency comb comparison is not trivial. To reduce the complexity of such an experiment, we propose to implement the experimental setup depicted in Fig. 2(a). A CW laser at 1560 nm and its second harmonic at 780 nm are used to characterize how well the frequency comb can transfer the stability of the 1560 nm CW laser to 780 nm by comparing to an optical doubling stage. PPLN-waveguide-based frequency doubling has been shown to support stability below 1×10^{-17} [48] at 1 s averaging time.

A CW laser oscillating at 1560 nm is sent through a PPLN doubling stage. The output of the PPLN contains both CW light at 1560 nm and 780 nm, with optical power reaching 50 mW and 1 mW, respectively. We then introduce the dual beam to the

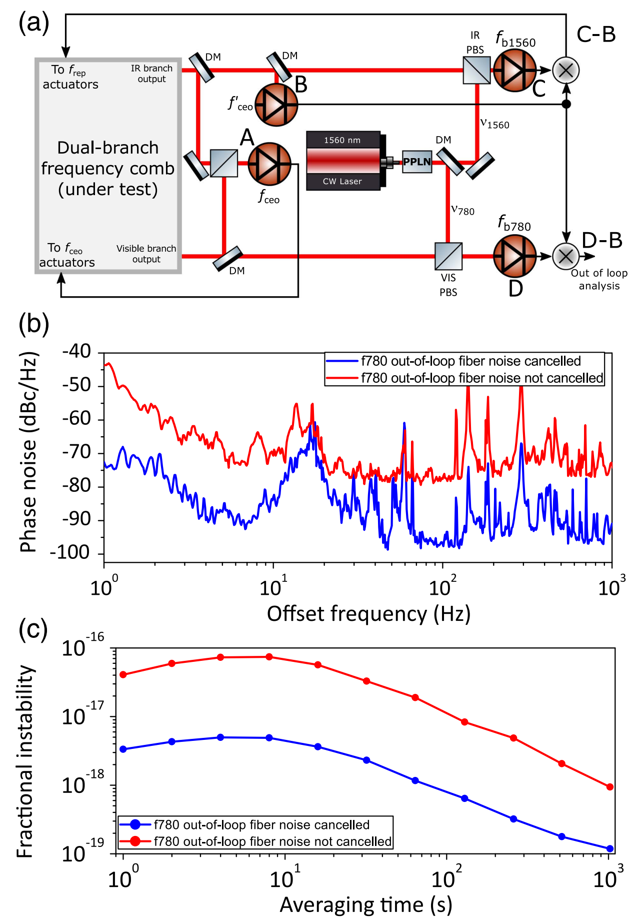

Fig. 2. (a) Experimental setup to evaluate the residual noise of the proposed method in this paper. A 1560 nm CW laser is used to stabilize the frequency comb. This laser is also doubled to 780 nm through a PPLN, and this doubled CW light is then compared to a comb mode at 780 nm. (b) Phase noise power spectral density of the setup in the non-fiber cancelled configuration (red) and fiber cancelled configuration (blue). (c) Fractional frequency instability in terms of Allan deviation of the setup in the non-fiber cancelled configuration (red) and fiber cancelled configuration (blue) (raw data acquired with a Λ-type frequency counter with 1 s gate time).

experimental setup shown in Fig. 1(b). To refer to the previous section [in particular, Fig. 1(b)], the 1560 nm light is combined with the IR branch through the IR PBS [in Fig. 2(a)] and beats on photodetector C providing the signal f_{b1560}. The 780 nm light is combined with the visible branch through the VIS PBS [in Fig. 2(a)] and beats on photodetector D providing the signal f_{b780}. A dichroic mirror spatially separates the two beams, and each beam beats with the frequency comb. The 1560 nm light beats with the IR, branch and the 780 nm beats with the visible branch. Dichroic mirrors are also used at the output of each branch to separate the 1064 nm light used to detect the dual-branch f_{ceo}. Additionally, the f_{ceo} signal generated in the IR branch is reflected by another dichroic mirror. Both f_{ceo} beatnotes exhibit SNR > 40 dB in 100 kHz resolution bandwidth. The free-space optical interferometers are all at least 50 cm long. Those interferometers include completely uncompensated non-common paths and inevitably introduce additive noise due to environmental disturbances, temperature fluctuations, etc. We phase lock the dual-branch f_{ceo} signal to an RF reference. As the 1560 nm CW laser is free running, we choose to phase lock the comb repetition rate to it so we can fully cancel the phase noise of the laser. The two optical beatnotes at 1560 nm and at 780 nm are depicted in Fig. 1(d). They both exhibit at least 30 dB SNR in 100 kHz resolution bandwidth, which is adequate for phase locking, frequency counting, and phase noise analysis. After detecting the beatnotes, we down mix them with the f'_{ceo} signal detected at 1100 nm from the IR branch to fully cancel the differential phase noise between both branches. Following the demonstration in the previous section, we can write

$$\begin{cases} \phi_C(t) - \phi_B(t) = n\phi_{\text{rep}}(t) - \phi_{\nu_{1560}}(t) \\ \phi_D(t) - \phi_B(t) = 2n\phi_{\text{rep}}(t) - \phi_{\nu_{780}}(t), \end{cases} \tag{7}$$

with ϕ_C the phase noise of the beatnote f_{b1560} between the 1560 nm CW light and the IR branch, ϕ_D the phase noise of the beatnote f_{b780} between the 780 nm CW light and the visible branch, $\phi_{\nu_{1560}}$ and $\phi_{\nu_{780}}$ are the phase noise of the CW light at 1560 nm and 780 nm, respectively, and n the mode number of the comb line beating with the 1560 nm CW light. By phase locking the RF signal carrying $\phi_C(t) - \phi_B(t)$ to an RF reference, we can thus write that

$$n\phi_{\text{rep}}(t) - \phi_{\nu_{1560}}(t) = 2n\phi_{\text{rep}}(t) - \phi_{\nu_{780}}(t). \tag{8}$$

As $\phi_{\nu_{780}} = 2 \times \phi_{\nu_{1560}}$, Eq. (8) highlights that this experimental setup aims at quantifying the similarity of the fractional frequency instability of the comb modes n and $2n$. However, one needs to take into account also the uncompensated noise quantities added in quadrature that will lead to the out-of-loop residual noise and instability.

The results of the experiment are presented in Figs. 2(b) and 2(c). We have first measured the phase PSD and the Allan deviation in a conventional dual-branch configuration by phase locking the f_{ceo} from the IR branch instead of the dual-branch one. The red curves in Fig. 2(b) and 2(c) show the residual error of such a setup. The phase noise at 1 Hz offset frequency is −45 dBc/Hz, and the instability level in terms of Allan deviation is 4×10^{-17} at 1 s averaging time and averages down to 1×10^{-18} at 1000 s (approximately $1 \times 10^{-16}/\sqrt{\tau}$). This noise level is typical from differential phase noise between optical branches based on optical amplification and spectral broadening. To observe that the proposed method efficiently cancels this differential phase noise, we now phase lock the dual-branch f_{ceo}. We observe a phase noise reduction to −70 dBc/Hz and an instability level at approximately 3×10^{-18} at 1 s averaging time and down to 1×10^{-19} at 1000 s (approximately $1 \times 10^{-17}/\sqrt{\tau}$). As mentioned previously, we suspect this excess noise to be due to uncompensated optical paths and interferometers as well as photodetectors. The measured instability is in the same range as what other groups working on the topic are measuring. It would be possible to minimize this noise by placing the interferometers in a vacuum chamber. However, this study does not focus on how to passively minimize the noise induced by the optical interferometers. We will explore in the next section the noise induced by electrical chains, expected to be lower than optical interferometers.

B. Ultimate Performance Characterization

In order to eliminate the noise added by optical uncompensated paths, interferometers, and photodetectors, we have implemented the experimental setup illustrated in Fig. 3(a). Following the idea of the previous section to evaluate the out-of-loop residual error, we now use a CW laser at 1064 nm. Light at 1064 nm is convenient, as it overlaps with the wavelength range we realized for the dual-branch f_{ceo} detection. Through a PBS, we combine the 1064 nm light with the two supercontinua between the PBS supercontinua and the photodetector. We can then detect all the signal on a single photodetector. The CW laser at 1064 nm beats with the IR branch as well as with the visible branch, and the two branches also interfere on the photodetector. We then completely overcome the noise added by the interferometer, as it is compensated by the phase-locked loop.

The photodetected RF signals are shown in Fig. 3(b). We can observe the f_{ceo} signal at around 60 MHz, the IR beatnote f_{IR} between the IR branch and the CW laser at around 40 MHz, and the visible beatnote f_{VIS} between the visible branch and the CW laser at around $f_{\text{ceo}} + f_{\text{IR}} = 100$ MHz. We split the output of the photodetector into three distinct electrical paths in order to isolate, filter, and amplify each signal. While mixing the optical beatnotes with the f_{ceo} signal from the IR would not be necessary here, as the mode numbers are equal, we still use the down mixing to evaluate as well the noise added by the mixers. The CW laser at 1064 nm being operated in free-running mode, we again phase lock the down-mixed signal to an RF reference. The dual-branch f_{ceo} signal is phase locked to an RF reference as well.

The noise and instability performance are presented in Figs. 3(c) and 3(d). Should RF amplification, filtering, and mixing be noiseless, the noise measured should be the quadrature-sum of the in-loop noise presented in the first section [see Figs. 1(e) and 1(f)]. However, using fairly low SNR will introduce noise in the electrical chain. While we have tried to use as short cables as possible, it is very challenging to overcome entirely uncompensated electrical noise. We can then observe a phase noise PSD higher than the one presented in the in-loop one. However, the phase noise level at 1 Hz offset frequency is < − 85 dBc/Hz. We observe that all the technical noise between 10 Hz and 1 kHz, due to acoustic noise and vibration, is cancelled, as we use only one interferometer, and accumulated phase noise in the IR and visible branches is correlated. The fractional instability is 6×10^{-19} at 1 s averaging down to 1×10^{-20} at 1000 s (this limit is discussed in Supplement 1). This measurement represents the current absolute and lower limits of our proposed method to compare optical frequencies from the IR to the visible.

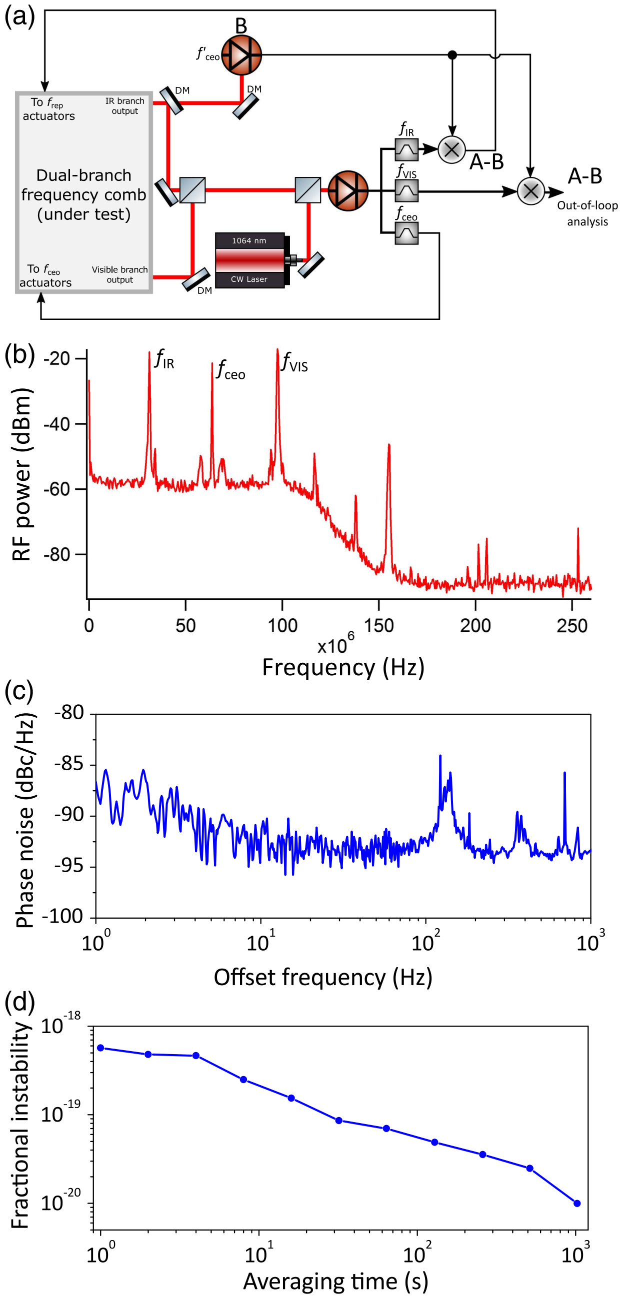

Fig. 3. (a) Experimental setup, to evaluate the residual noise of the proposed method, overcoming the excess noise from non-common path optical interferometers. A 1064 nm CW laser beats with the stitched visible and infrared branch on a common photodetector as well as with the dual-branch f_{ceo} detection. (b) RF spectrum of the signal photodetected exhibiting the two optical beatnotes f_{IR} and f_{VIS}, and f_{ceo}. Resolution bandwidth is 100 kHz. (c) Phase noise power spectral density of the out-of-loop beatnote f_{VIS}. (d) Fractional frequency instability in terms of Allan deviation of the out-of-loop beatnote f_{VIS}. This instability level reveals the lower limit of the proposed method (raw data acquired with a Λ-type frequency counter with 1 s gate time).

Indeed, uncompensated optical paths can always be better engineered, thermally and acoustically isolated, and made vibration insensitive by using dedicated vibration isolators. Note that the detection of the IR f_{ceo} signal is cancelled out, as the mode numbers are equal. Nevertheless, this free-space detection is kept shorter than about 50 cm. It is worthwhile to notice as well that, practically, with two distinct CW lasers, such detection used in this setup could be possible only for wavelength that can share the optical bandwidth of an interference filter of a photodetector. This section aimed at showing the noise that a free-space optical interferometer can induce, and it will ultimately depends on environmental conditions in different laboratories.

Other sources of noise should also be addressed. Indeed, wavelength-dependent higher-order noise will occur through the supercontinuum generation due to the refractive index, as well as the intensity noise on the pulse train that can be converted into phase noise. These sources of noise will be extremely complicated to measure although will need to be addressed for the next generation of optical frequency comb. However, the phase noise and instability measured here surpass by more than one order of magnitude the best current optical local oscillator and optical clock.

Finally, our experimental setup uses a Rubidium clock as a frequency reference. This reference at 10 MHz has a fractional frequency uncertainty of 10^{-12}; scaled to an optical frequency, this corresponds to an uncertainty of about 10^{-19}. We could not observe any frequency offset due to the spectral purity transfer within this uncertainty. Further studies will be devoted in evaluating thoroughly the inaccuracy of the method we propose in this paper.

4. CONCLUSIONS

To summarize, we have developed, implemented, and characterized a new method to fully compensate the phase noise accumulated in a dual-branch frequency comb system spanning 500 nm to 2200 nm. We believe that this method will be useful for the synthesis and comparison of optical frequencies as well as stability transfer across the full optical span. We have demonstrated that Er:fiber laser technology used in this study surpasses any competitive technology with residual errors lower than 1×10^{-19} level due to the combination of a low jitter oscillator and high-bandwidth actuators. Using the reported new technique, we have shown with an out-of-loop measurement that the performance of the stability transfer from 1560 nm to 780 nm is not jeopardized by the frequency comb to 3×10^{-18} at 1 s averaging time and 1×10^{-19} at 1000 s. This is limited mainly by uncompensated optical paths. Moreover, we have measured the lower limit of the method to be at an instability level of 6×10^{-19} at 1 s and 1×10^{-20} at 1000 s.

Finally, we have demonstrated for the first time that this scheme is the only current way to fully cancel differential phase noise to access wavelengths between IR and visible spectral regions at this instability level. As a result, any optical transition in the visible can now benefit from state-of-the-art optical local oscillators, which will ultimately impact the fields of applied and fundamental physics.

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See Supplement 1 for supporting content.

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